Project Design Phase-II / Requirement Analysis

Data Flow Diagram & User Stories

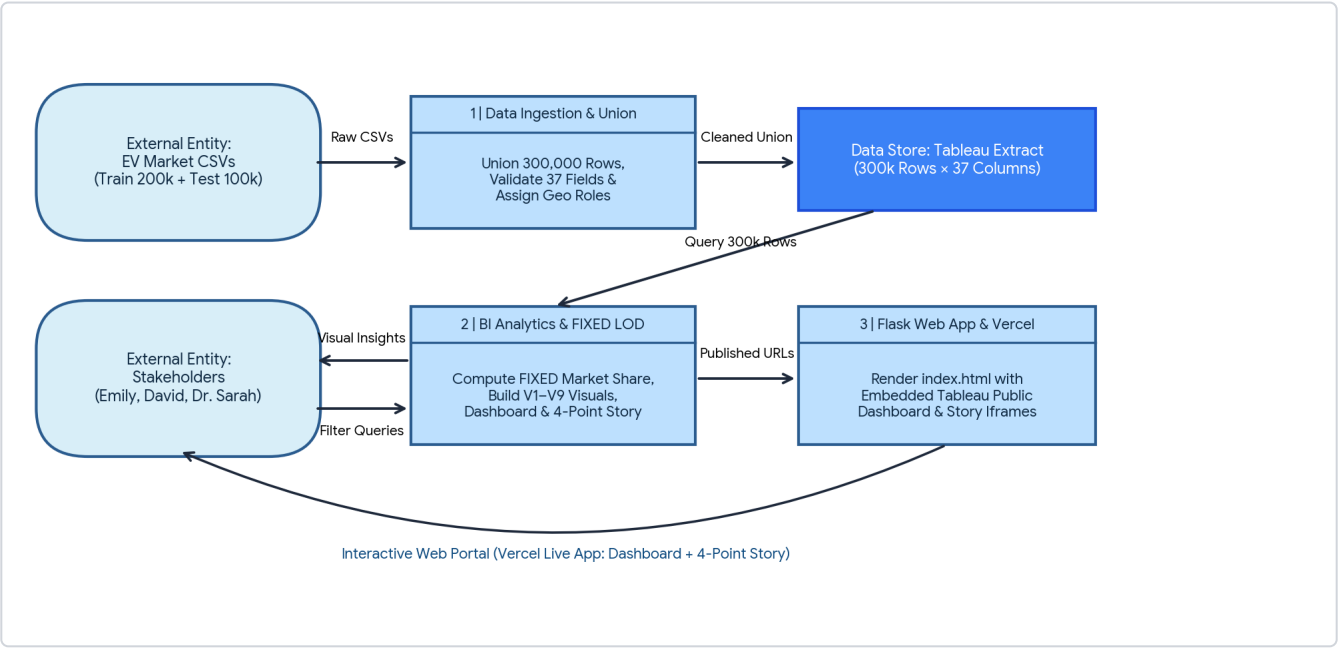
Date	28 October 2026
Team ID	Ujjawal Rathod (Individual Capstone)
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Data Flow Diagram (DFD): Visual representation of how the 300,000 EV market records flow from raw CSV files through Tableau Union & FIXED LOD processing into the interactive Tableau Public views and Flask/Vercel web application.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	Source or destination of data outside the system (EV Market CSV Files; Stakeholders Emily, David, Dr. Sarah).
Rectangle with numbered header	Process	Analytical or web activity that transforms data (1: Data Union & Prep, 2: BI Analytics & FIXED LOD, 3: Flask Web App).
Solid Blue Rectangle	Data Store	Repository storing the unified 300,000-row × 37-column Tableau Data Extract (`.twbx`).
Labeled Arrow	Data Flow	Movement of raw records, queries, filter actions, and embedded visual outputs across components.

Project Data Flow Diagram (EV Market Tableau & Flask BI Architecture)



User Stories

User Type	Functional Requirement (Epic)	USN	User Story / Task	Acceptance Criteria	Priority	Release
Emily Johnson (Strategy Director)	Category Sales & Volume Analysis	**USN-1**	As a Market Strategy Director, I want to view total annual sales across all 15 EV vehicle categories sorted descending (V1) so that I can identify which segments drive the highest volume.	Horizontal bar chart (V1) displays Two-Wheeler Scooter leading, followed by Sedan, Hatchback, and Compact SUV.	High	Sprint-1
Emily Johnson (Strategy Director)	Historical Trend Analysis	**USN-2**	As a Market Strategy Director, I want to track annual sales trends from 2018 to 2026 for the Top 5 categories (V3) so that I can evaluate whether segments are growing or stabilizing.	Multi-line chart (V3) displays discrete model years 2018–2026 for the top 5 categories with clean Millions (M) axis formatting.	High	Sprint-1
David Chen (BI Manager)	Competitor Market Share (FIXED LOD)	**USN-3**	As an Automotive BI Manager, I want to see the Top 10 manufacturers with their true global market share across all 200 competitors (V2) without Top-N filter distortion.	Bar chart (V2) uses `SUM([Sales])/MIN({FIXED : SUM([Sales])})` showing Ampere Works at ~0.59% share.	High	Sprint-1
David Chen (BI Manager)	Battery Technology Share	**USN-4**	As an Automotive BI Manager, I want to analyze the percentage share of sales across all 5 battery chemistries (V4) so that I can track dominant vs. emerging	Donut chart (V4) shows NMC (33.7%) and LFP (32.0%) leading, followed by NCA (17.3%), Sodium-Ion (9.9%), and Solid-State (~7%).	High	Sprint-1

User Type	Functional Requirement (Epic)	USN	User Story / Task	Acceptance Criteria	Priority	Release
			battery technologies.			
David Chen (BI Manager)	Geographic Production Mapping	**USN-5**	As an Automotive BI Manager, I want an interactive filled map of manufacturing countries (V7) showing total sales and distinct vehicle counts so that I can benchmark production hubs.	Filled map (V7) shades all 10 manufacturing countries (China ~237M, India, Vietnam leading) with Distinct Vehicle ID count in tooltips.	High	Sprint-2
David Chen (BI Manager)	Regional Product Positioning	**USN-6**	As an Automotive BI Manager, I want a Region × Vehicle Category sales heatmap (V9) across all 7 market regions so that I can spot regional expansion opportunities.	7×15 square heatmap (V9) highlights category strength per region (e.g., Crossover in Asia Pacific at 29.7M) totaling 2,323.3M units.	High	Sprint-2
Dr. Sarah Williams (Researcher)	Driving Range & Efficiency	**USN-7**	As a Sustainable Mobility Researcher, I want to compare average driving range (km) and energy efficiency (Wh/km) across all 15 categories (V5) against the overall market average.	Bar chart (V5) shows Luxury Sedan (~550 km) down to Two-Wheeler Scooter (~105 km), colored by Avg Efficiency (Wh/km) with an average line.	High	Sprint-2
Dr. Sarah Williams (Researcher)	Battery Range Evolution	**USN-8**	As a Sustainable Mobility Researcher, I want to evaluate average driving range trends from 2018 to 2026 by battery type (V6) on a non-misleading zero-anchored scale.	Line chart (V6) uses a custom 0–400 km axis proving average range stays steady (~324 km) across all battery chemistries.	High	Sprint-2
		USN-9	As a Sustainable Mobility	Bar chart (V8) anchored at 0–5	Medium	Sprint-2

User Type	Functional Requirement (Epic)	USN	User Story / Task	Acceptance Criteria	Priority	Release
Dr. Sarah Williams (Researcher)	Customer Satisfaction Benchmarking		Researcher, I want to compare average customer ratings across all 15 vehicle categories (V8) on a full 0–5 scale with an overall reference line.	confirms consistent satisfaction (~4.08 to 4.09 average rating) across all categories.		
All Stakeholders (Dashboard Users)	Interactive Cross-Filtering	**USN-10**	As a dashboard user, I want global dropdown filters (Market Region, Battery Type, Model Year) and click-to-filter actions on V1, V4, and V7 so all 9 visuals update together.	Changing any global filter or clicking a country/battery slice updates all 9 worksheets within 1–2 seconds.	High	Sprint-3
Executive Viewer (Web User)	Persona Story Navigation	**USN-11**	As an executive viewer, I want a 4-point Tableau Story (Overview, Emily Strategy, David Competition, Dr. Sarah Research) with curated insight callouts.	Story navigator switches cleanly across all 4 persona dashboards with zero overlapping legends or clipped charts.	High	Sprint-3
BI Administrator (Ujjawal Rathod)	Flask Web App & Vercel Deployment	**USN-12**	As a BI Administrator, I want to embed the live Tableau Public Dashboard and Story into a Python Flask web application deployed on Vercel.	Flask route (`'/'`) renders `index.html` with responsive CSS and live Tableau iframes accessible via a public `.vercel.app` URL.	Medium	Sprint-3